

# Olivia Jackson Lambert

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## Education

### Master of Information & Data Science

University of California, Berkeley | Anticipated Graduation: 05/2027

Student Representative, Relevant Coursework: Machine Learning, Natural Language Processing, Deep Learning, Data Engineering

### Master of Physics, Astrophysics | First Class with Honours (4.0 GPA Equivalent)

University of Edinburgh | 2017 - 2022

President - Physics & Astronomy Society, Edinburgh Award for Leadership, Carnegie Scholarship Recipient, UKSEDS Diversity Champion

## Technical Skills

**Programming Languages:** Python, R, SQL, C++, LaTeX

**Data Science & ML:** scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, Prophet, statsmodels, pandas, NumPy, SciPy

**Data Engineering:** Airflow, DBT, Spark, PySpark, ETL Pipelines, Docker, Git

**Cloud & Databases:** AWS (CCP Certified), GCP, Snowflake, Redshift, PostgreSQL, BigQuery, Neo4j, MongoDB, NoSQL Databases

**Visualization & BI:** Streamlit, Looker, Matplotlib, Tableau, ggplot

**Techniques:** Regression, Hypothesis Testing, Statistical Inference, A/B Testing, Experimental Design

## Relevant Experience

### Analytics Engineer

Democratic National Committee | September 2024 - November 2024

- Deployed machine learning models serving 100K+ monthly users by building SQL models in DBT and orchestrating DAGs in Airflow
- Executed gradient-boosted XGBoost model on 160M+ voter records to predict donor propensity, optimizing fundraising campaigns
- Refactored data delivery pipeline by migrating to DBT and GCP, reducing API load by 99.999% and eliminating throttling issues
- Developed real-time election night dashboards in Looker, delivering insights to senior stakeholders before major media outlets

### Data Scientist - Consultant

Kenvue - Johnson & Johnson | September 2023 - March 2024

- Built global demand forecasting pipeline using Prophet and PySpark, generating 10K+ monthly time series forecasts across 5 continents
- Developed supervised learning classification module with sktime and scikit-learn to categorize time series, automating a previously manual process and saving the client 100s of annual work hours and 1000s of dollars a year
- Productionized ETL pipeline with Redshift, Snowflake, and Streamlit, creating automated SQL-based interactive dashboard deliverables

### Data Scientist - Consultant

Fidera Group | March 2023 - September 2023

- Architected automated ETL workflows with Python and Airflow to integrate datasets, improving data accessibility for stakeholders
- Engineered data validation tests and introduced Git version control and Docker containerization for improved quality control

## Research Experience

### Research Assistant - Computational Astrophysics

Institute for Astronomy | May 2020 - September 2020

- Extracted and analyzed hundreds of FITS images from Sloan Digital Sky Survey using advanced SQL queries and C++ algorithms
- Built automated data pipelines in Python with Astropy, SciPy, pandas, and Matplotlib for feature extraction and visualization

### Research Assistant - Computer Vision & Machine Learning

National Physical Laboratory | May 2019 - September 2019

- Developed autonomous sky imaging software for Raspberry Pi devices, implementing dynamic exposure adjustment algorithms
- Applied machine learning with OpenCV to classify cloud cover patterns, improving satellite re-entry prediction accuracy by 23%

## Technical Projects

### Receipt Entity Extraction with Transformer NLP

- Built multi-task transformer pipeline using Hugging Face and PyTorch to classify and extract fields from retail receipts
- Technologies: Transformers, Named Entity Recognition (NER), Multi-Task Learning, CUDA, Tokenizers

### High Energy Particle CNN Classifier

- Developed convolutional neural network to predict particle types from detector tracks, achieving 93% accuracy
- Technologies: Deep Learning, Hyperparameter Optimization, TensorFlow, Keras, scikit-learn

## Certifications & Publications

**Certifications:** AWS Certified Cloud Practitioner (CCP) | Certified Associate Python Programmer (PCAP) | DBT Fundamentals

**Publication:** "The Need for Large Sample Numbers to Demonstrate that Martian Environments Are Lifeless" - Nature Astronomy (12/2024)